System One's Last Exam

Benchmarking Generalist Competence and Decision Reliability

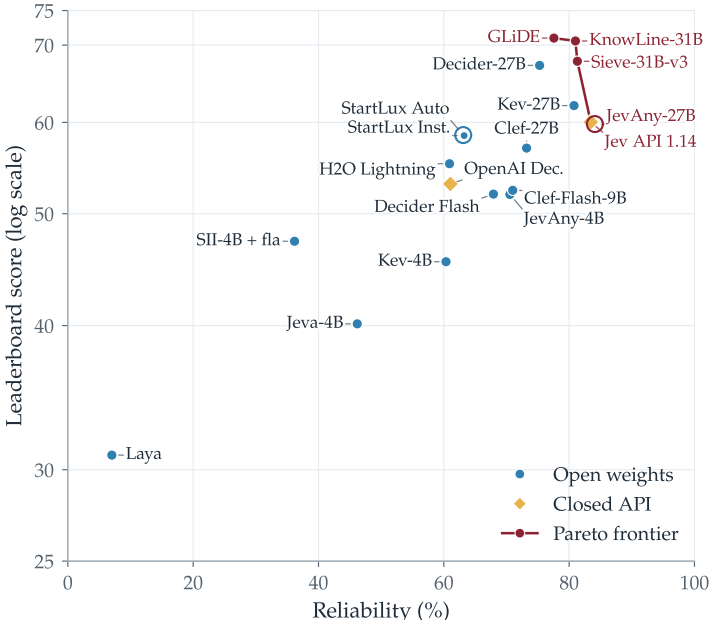
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System One models promise fast, general-purpose decision-making without requiring lengthy chain-of-thought generation. However, evaluating these models requires understanding not only the breadth of their competence but also the reliability of their judgments. Existing benchmark scores can be difficult to interpret due to ambiguous annotations, inconsistent task criteria, and limited assessments of decision reliability. We introduce **System One's Last Exam**, a comprehensive benchmark that jointly evaluates **generalist competence and decision reliability** across eleven domains and five capability families. Our benchmark combines carefully curated existing datasets with purpose-built diagnostic tasks, featuring explicit task rules, verifiable reference answers, and structured difficulty tiers. Beyond conventional accuracy, we assess decision reliability through a unified metric that captures consistency under presentation changes and the ability to recognize when no valid answer is available. We further include controlled execution-history tasks to examine whether models can distinguish successful outcomes from violations of procedural requirements. Evaluations across 21 model configurations reveal that the benefits of model scaling and additional reasoning vary substantially across tasks, suggesting that neither provides a universal solution to improving decision competence. More importantly, strong overall competence does not necessarily imply reliable judgment: the most reliable configuration ranks only ninth in overall performance. Together, these findings reveal important differences in how System One models acquire competence and maintain decision reliability, providing a systematic foundation for understanding their capabilities and limitations.

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(a) Deep-dive ranking



(b) Accuracy and reliability

Figure 1: **S1-Bench at a glance.** (a) Leaderboard scores for 21 model configurations; greater depth indicates a higher score, while horizontal placement is illustrative. (b) Scores versus reliability for the 19 configurations with reliability measurements; the red line marks the Pareto frontier.

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1. Introduction

System Two reasoning has substantially advanced language model capabilities in mathematics, programming, and science [2, 20, 30, 52]. Many software workflows also require models to make repeated decisions over predefined outcomes, such as routing requests, selecting tool actions, and verifying task completion. System One models, exemplified by Jev [1], predict a probability distribution over admissible outcomes in a single forward pass. This design reduces inference latency by avoiding autoregressive output generation and produces structured decisions that downstream software can use directly.

JevBench provides a benchmark for typed decisions [4], while System1-Arena compares System One models with generative language models on both static decisions and closed-loop tasks [9]. Building on these evaluations, we argue that a benchmark designed to characterize the capabilities and reliability of System One models should meet three requirements. It should *measure performance as task complexity increases within each domain*, *evaluate the model both as an actor and as a verifier*, and *test whether decisions respond appropriately to changes in the available options*.

We introduce **System One's Last Exam** (S1-BENCH), a benchmark of 2,919 tasks across eleven domains in five capability families, covering semantic understanding, knowledge and reasoning, safety and policy compliance, evidence assessment, and agent task completion. The benchmark uses a common decision interface and scores performance at the level of the original task. Its design follows the three requirements above.

Difficulty within domains. An aggregate score can obscure weaknesses on demanding tasks when routine cases account for much of the evaluation. We stratify tasks in nine domains using documented, manually specified criteria for task complexity. In web interaction, for example, the tiers distinguish single-step decisions from routes that require a sequence of actions across successive states. We report performance at each tier and construct the main leaderboard by weighting all eleven domains equally, using the hardest tier in each tiered domain and the domain score in each of the two untiered domains.

Action selection and completion verification. Agent benchmarks assess functional task success and can also account for intermediate milestones [49, 85]. We separately evaluate whether a model can execute a task and whether it can verify a recorded execution. In tool use and web interaction, the model selects each action in a simulator or along a replay-verified web route, so its decisions determine the states it subsequently encounters. For completion verification, we include controlled pairs of executions that reach the same final state but differ in whether they satisfy an intermediate requirement. By holding the final state fixed while varying the correct verdict, these pairs test whether the verifier uses evidence from the execution history. Each task has traceable provenance, and evaluation references are kept separate from model-visible inputs.

Reliability under changes to the option set. Related work on probabilistic coherence evaluates how predictions change under transformations of the options [25]. We focus on two interventions with distinct expectations. Reordering the options preserves the correct outcome, so the selected outcome should remain stable. Removing the correct option makes the original question unanswerable with the remaining options, so the model should recognize that none is correct. We combine these tests into a reliability score and report it separately from the main leaderboard.

Across 21 model configurations, mean scores decline with difficulty in all nine tiered domains, with the largest declines in agentic tasks. In web interaction, the mean score falls from 93.0 on single-step tasks to 33.3 on hard routes. The two scores also distinguish different configurations. JevAny-27B ranks first among the 19 configurations assessed for reliability and ninth on the main leaderboard.

Our contributions are threefold.

- **A cross-domain benchmark with explicit difficulty criteria.** We provide a common task-level evaluation across eleven domains and characterize performance as task complexity increases within each tiered domain.
- **An agentic evaluation that separates execution from verification.** We evaluate action selection over complete episodes and use controlled execution pairs to test whether completion verdicts reflect requirements that cannot be inferred from the final state alone.
- **A comparative study of capability and reliability.** We evaluate 21 model configurations and use controlled option interventions to measure decision stability and recognition of unanswerable choices alongside task performance.

2. Background: System One Decision Models

We use *System One decision models* to denote models that map a supplied state and a decision specification to structured predictions over admissible outcomes. This terminology follows the Jev interface and subsequent decision-model research [1, 23]. It identifies an input–output paradigm rather than a particular architecture or an equivalence to human cognition.

Following existing decision-task formulations [23], we represent an instance as

$$x = (s, q, \mathcal{C}), \tag{1}$$

where s contains the supplied information, q specifies the decision objective, and $\mathcal{C} = \{c_1, \ldots, c_K\}$ defines the admissible outcomes and their descriptions. The supplied state may contain text, structured records, retrieved passages, or interaction histories. It is the model-visible representation, not necessarily the complete state of an underlying environment.

A decision model returns a distribution $p_\theta(\cdot \mid x)$ satisfying

$$p_\theta(c_k \mid x) \geq 0, \qquad \sum_{k=1}^{K} p_\theta(c_k \mid x) = 1. \tag{2}$$

This distribution can be obtained through a learned decision head or a readout from language-model logits, among other implementations [28, 54]. Our formulation does not require models to share an internal architecture.

We consider three output formats supported by the Jev-style interface [73]:

- **Categorical selection (`choice`).** The outcomes are explicitly specified alternatives without an imposed ordinal structure. The predicted outcome is

$$\hat{y}(x) = \arg\max_{c \in \mathcal{C}} p_\theta(c \mid x). \tag{3}$$

- **Binary judgment (`noul`, the interface's yes/no question type).** The outcome space is $\{0, 1\}$, and the model returns $p_\theta(1 \mid x)$, interpreted as the probability assigned to the affirmative outcome.

- **Ordinal assessment (`score`).** The outcomes are ordered descriptive levels with grades $g(c_k)$ on the source's native scale. The predicted grade is that of the most probable level,

$$\hat{g}(x) = g\big(\hat{y}(x)\big). \tag{4}$$

  An ordinal ordering alone does not establish equal semantic distances between levels, so we compare grades only through a bounded credit for adjacent levels (Section 3.3).

These formats describe the output space. They do not determine the task domain or the reasoning required to answer a question.

Advantages of the decision interface. The same interface supports classification, routing, ranking, and verification by changing the supplied state and decision specification. Explicit outcome spaces produce structured outputs that downstream systems can consume directly, without extracting an answer from free-form text. Probability readouts also support comparing alternatives and applying application-specific decision thresholds. Implementations with direct prediction heads or answer-token readouts can avoid generating extended reasoning traces, reducing the decoding work needed for a decision [28, 54, 76]. These are interface and implementation advantages; they do not by themselves establish that a model's decisions are correct or robust.

3. Benchmark Design and Evaluation

System One's Last Exam comprises **2,919** natural tasks in **eleven** domains, grouped into five capability families (Table 1 and Figure 3). All tasks use the decision interface of Section 2: 2,300 are static tasks, 337 are web tasks (100 single-step judgments and 237 reference-route tasks), and 282 are interactive tool-use episodes. Figure 2 outlines how tasks are constructed, posed to models, and scored. The scored unit is always the natural task, such as a review, a conversation, a query–passage pair, or an episode, however many decisions it contains. Per-source construction details appear in Appendix A.

3.1. Sources and Conversion

Source fidelity. Most items come from public datasets with reliable labels or executable evaluators; the rest are built in public environments, such as the controlled VirtualHome trajectories and new BrowseComp-Plus subgoals. Each item records its source, pinned version, and original identifier. Conversion never rewrites labels, and items with doubtful labels are held out; the one exception is 18 references corrected after evaluation (Appendix A.6). Model inputs exclude evaluation references, oracle states, and construction annotations.

Conversion into decisions. Multiple-choice items keep their options. Open-answer math problems receive distractors drawn mainly from the wrong answers of a base model answering without reasoning. A multi-label task becomes one independent `noul` question per native label, never a choice among preassembled label sets. Graded judgments keep their native scales, and agent steps become menus (Section 3.3). Tasks that require vision or free-text output are excluded.

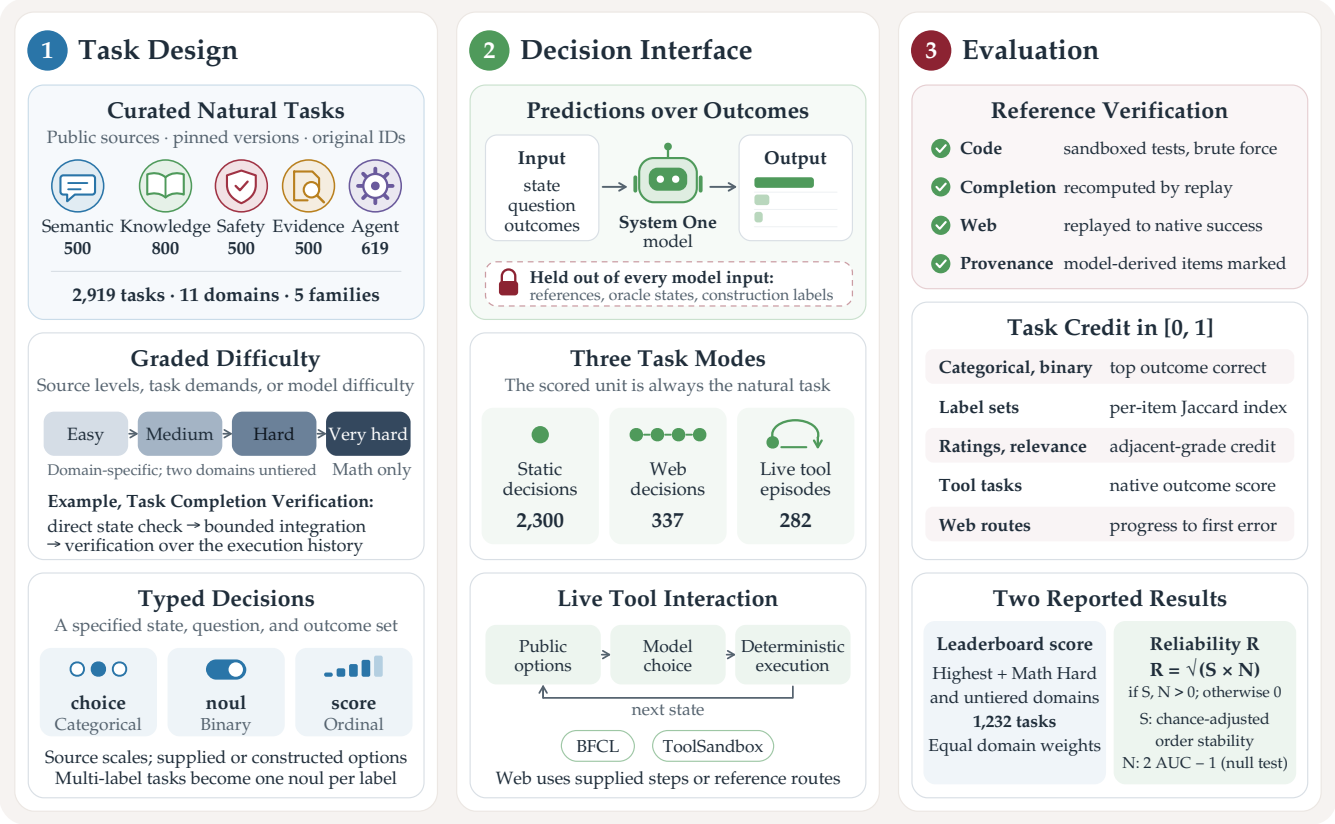


Figure 2: **Overview of System One's Last Exam.** Curated tasks are expressed as typed decisions, with domain-specific tiers where applicable. Static tasks may contain multiple questions. The execution loop applies to live tool episodes. Task credits yield the leaderboard score; separate order and null tests measure reliability.

3.2. Graded Difficulty

Each tiered domain grades difficulty by its own documented criterion, and tiers are not calibrated across domains. No criterion uses input length, option count, or artifacts of our conversion. Sentiment & Emotion Understanding and Knowledge & Commonsense are untiered, because their sources share no difficulty scale.

Tier criteria. Scientific Reasoning and Math follow their sources: the benchmark of origin, the MATH level, and the number of calculation steps in GSM8K. Intent & Routing, Retrieval & Reranking, Safety, Task Completion Verification, Tool & API Use, and Text-based Web Interaction follow task features, such as the number of intents to identify; in the 20 hard VirtualHome trajectories [44, 61] of Task Completion Verification, the answer depends on what happened during execution, not only on the final state. Code Understanding follows how hard the given solution is to judge rather than LeetCode's problem levels, which did not order judging accuracy. Appendix A gives the criterion of each domain.

3.3. Executable and Auditable Evaluation

Agentic decision sequences. Rather than generating actions, the model answers one `choice` request per step. The state holds the goal, the current observation, and the full public history, and the options come

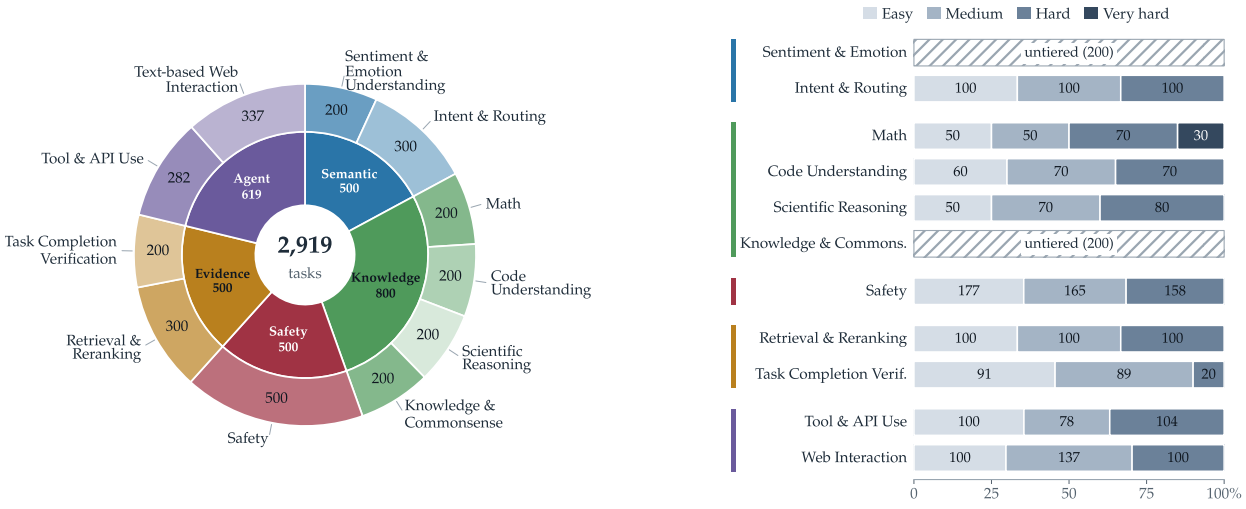
Table 1: **Composition of System One's Last Exam.** Counts are natural tasks. Tiers give easy/medium/hard counts; Math additionally has 30 very hard AIME problems, and "–" marks domains without tiers. Where tiers draw on different sources, sources are listed from easy to hard, separated by →. "Seq." denotes a sequence of `choice` requests.

Domain	Tasks	Tiers	Format	Sources
Semantic understanding				
Sentiment & Emotion Understanding	200	–	`choice`, `score`	Amazon Reviews 2023 [37], GoEmotions [21], SST-5 [65], DynaSent [60], ARTS [78]
Intent & Routing	300	100/100/100	`choice`, `noul`	CLINC150 [42], BANKING77 [6], SGD [62] → NatCS [31] → NLU++ [7]
Knowledge and reasoning				
Math	200	50/50/70	`choice`	GSM8K-Platinum [15, 74], MATH500 [35, 48], AIME 2025–2026 [3]
Code Understanding	200	60/70/70	`noul`	CodeJudgeBench [39], LiveCodeBench [38], LeetCodeDataset [77]
Scientific Reasoning	200	50/70/80	`choice`	ARC-Challenge [12] → MMLU-Redux [27] → GPQA Diamond [63]
Knowledge & Commonsense	200	–	`choice`	MMLU-Redux [27, 34], HellaSwag [82], CommonsenseQA [70]
Safety and policy compliance				
Safety	500	177/165/158	`choice`, `noul`	RAGTruth [51], ReasoningShield [43], HarmThoughts [40], DynaBench [36], Aegis 2.0 [29], OR-Bench [19]
Evidence assessment				
Retrieval & Reranking	300	100/100/100	`noul`, `score`	TREC DL [17, 18] → BRIGHT [68], Touché 2020 [5, 72] → ACORD [75]
Task Completion Verification	200	91/89/20	`choice`, `noul`	BrowseComp-Plus [10], CALVIN [50], τ-bench [79], MiniGrid [11], EAI-VirtualHome [44, 61]
Agent task completion				
Tool & API Use	282	100/78/104	live seq.	BFCL [55] → ToolSandbox [49] → BFCL
Text-based Web Interaction	337	100/137/100	`choice`, `noul`; seq.	Mind2Web [22] → BreakingWeb [81]

only from public information: documented functions or visible controls, argument values and text spans grounded in the request, the page, or earlier results, and an option to finish. Tool tasks run live in the BFCL and ToolSandbox simulators under budgets on decisions and tool calls: deterministic code executes each choice without any auxiliary model, a call that lacks a grounded required argument is reported back to the model rather than completed with an invented value, and the native checkers accept any route that reaches the required state. Medium and hard web tasks are teacher-forced along a reference route that the unchanged BreakingWeb checker accepts, so no browser is needed. They are scored by *progress*, the fraction of scored decisions answered correctly before the first error, and replay-verified alternative orders and harmless detours are accepted.

Reference verification. Code labels are verified by sandboxed execution (Appendix A.2), MMLU-Redux items are restricted to its error-free annotations, executable completion references are recomputed by replay, and web reference routes are replayed to native success. The 150 tasks with model-derived references are marked in the item metadata.

Scoring. Each natural task receives a credit in $[0, 1]$ (Figure 2): exact correctness of the top-probability outcome for categorical and binary decisions; the per-item Jaccard index for label sets; the adjacent-grade credit $\max(0, 1 - |\hat{g} - g|/2)$ for ratings and graded relevance, where $\hat{g}$ is the predicted grade (Equation (4)) and g the reference grade; the native outcome score for tool tasks; and progress for multi-step web tasks. The only static tasks with several questions are the label sets of Intent & Routing, one `noul` question per native label; the predicted set contains the labels whose affirmative probability exceeds one half. Refusals and over-length inputs count as failures, and inputs are never truncated: a static task or tool episode with an explicit refusal receives zero credit, a refused web decision counts as wrong and ends route progress, and a task with any request longer than the model's native context window receives zero credit. Appendix B lists every such case.



(a) Families and domains **(b) Difficulty tiers within each domain**

Figure 3: **Composition of the benchmark.** (a) Natural tasks by capability family (inner ring) and domain (outer ring). (b) Share of each domain's tasks in each difficulty tier, with task counts on the bars; side rules mark the families in (a). Tiers are defined within each domain and are not calibrated across domains.

3.4. Competence and Reliability Metrics

Capability: performance across domains. Let $c_i(m) \in [0,1]$ be model m's credit on natural task i, computed using the task-specific rules above. For domain d, let $\mathcal{T}_d$ contain its highest difficulty tier, except that Math includes both hard and very hard tasks; for domains without tiers, it contains all tasks. The leaderboard score is

$$C(m) = \frac{100}{D} \sum_{d=1}^{D} \frac{1}{|\mathcal{T}_d|} \sum_{i \in \mathcal{T}_d} c_i(m), \qquad D = 11. \tag{5}$$

Thus, tasks have equal weight within each domain, and domains have equal weight regardless of their size or the number of model requests per task. This selection contains 1,232 natural tasks: the hard and very hard tiers of Math (70 and 30 tasks, respectively), the hard tier of each other tiered domain, and all tasks in the two untiered domains. We focus the leaderboard on the upper tiers because the lower tiers separate models poorly: across the 21 configurations, the average easy-tier score exceeds 85 in eight of the nine tiered domains. Results for the remaining tiers are reported separately to show how performance changes with difficulty.

Reliability: stability under option reordering. An unchanged question should not elicit a different substantive answer merely because its options move. We test 939 static multiple-choice items with at least four options from seven domains,[1] excluding items with an option that refers to other options. Each item is presented in its original order and in 12 distinct shuffled orders; the state, question, and option texts are unchanged. Predictions are mapped back to the original option identities before comparison. Let $a_{ir}(m)$ be the agreement between the original prediction and shuffled run r for item i. Ties among top-probability options are split uniformly; failed or refused predictions receive zero agreement. The mean agreement A and chance-adjusted stability S are

$$A(m) = \frac{1}{12M} \sum_{i=1}^{M} \sum_{r=1}^{12} a_{ir}(m), \quad p_e = \frac{1}{M} \sum_{i=1}^{M} \frac{1}{K_i}, \quad S(m) = \frac{A(m) - p_e}{1 - p_e}, \tag{6}$$

where $M = 939$ and K_i is the number of options for item i. Here, p_e is the agreement expected from independent uniform choices. Stability is 1 for complete agreement and 0 at this chance baseline; it does not require the stable answer to be correct.

Reliability: recognizing when no option is correct. This test uses 597 items from Math, Scientific Reasoning, and Knowledge & Commonsense, domains in which no remaining option can still be acceptable once the correct one is removed, as a subjective label could be. For each item, we construct two versions: one retains the original options, and the other removes the sole correct option and relabels the remaining positions without gaps. Both versions ask whether at least one offered option correctly answers the question, using the same binary `noul` interface for every model. Define $p_{\text{null}} = 1 - p(\text{yes})$. A model should assign higher p_{null} when the correct option is absent. We measure separation using the area under the ROC curve (AUC), with the answer-removed versions as positives and the original versions as negatives, and define

$$N(m) = 2\ \text{AUC}(m) - 1. \tag{7}$$

[1] Math 200, Scientific Reasoning 195, Knowledge & Commonsense 192, Safety 137, Task Completion Verification 98, Sentiment & Emotion Understanding 70, and Intent & Routing 47.

The reported AUC pools both versions of all 597 items (1,194 requests). This is a pooled discrimination measure, not the fraction of individual pairs ordered correctly; AUC ties receive half credit. Perfect separation gives $N = 1$, while chance gives $N = 0$.

Combining the reliability tests. We require both stability and null discrimination to exceed chance:

$$R(m) = \begin{cases} \sqrt{S(m)N(m)}, & S(m) > 0 \text{ and } N(m) > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

The geometric mean penalizes weakness in either component: consistently choosing the same answer alone is insufficient. We display $100R$ alongside C in Figure 1. The reliability tests use separate, fixed subsets rather than the leaderboard's selected tiers, and they do not change the membership or scoring of the main benchmark.

4. Experiments

We report three properties of the 21 configurations (Section 4.1): competence across domains and tiers (Section 4.2), latency on one GPU (Section 4.3), and reliability under the two tests of Section 3.4 (Section 4.4). Open-weight configurations that answer without reasoning average 6.0 to 265 ms per task, and the two reliability tests order the configurations differently from each other and from the leaderboard.

4.1. Evaluation Setup

We evaluate 21 configurations (Table 2): 17 open-weight System One configurations with 0.4B to 31B parameters and four closed APIs. Two of the APIs, Jev API 1.14 and OpenAI Decisions, are System One models; GPT-6 Luna and Sol reason before answering and serve as references. Each configuration receives every task through its documented decision interface and runs in the settings documented by its developer; we change neither prompts nor settings, with two exceptions described in Appendix C. The open-weight configurations and the two System One APIs return a probability for every option, whereas GPT-6 Luna and Sol return a single answer without option probabilities. The reliability tests cover the 19 configurations other than GPT-6 Luna and Sol, and latency is measured for the 17 open-weight configurations (Section 4.3). Appendix C lists the release and settings of every configuration.

4.2. Main Results

Table 3 reports every model's score by domain and tier. Two configurations are affected by the zero-credit rule (Appendix B): 30 web tasks exceed Laya's 8,192-token context window, and OpenAI Decisions refuses 4 static tasks, 12 tool episodes, and 6 web decisions. Reliability is available for 19 configurations; GPT-6 Luna and Sol have no runs for these tests and are omitted from the reliability panel rather than assigned zero. Across the 21 configurations, the average score falls with each tier in all nine tiered domains.

4.3. Latency

Open-weight System One configurations that answer without reasoning average 6.0 to 265 ms per task on one GPU, and none of them that is slower than GLiDE (113 ms) has a higher leaderboard score (Figure 4).

Table 2: **The 21 evaluated configurations,** in the order of Table 3 within each access type. Size: parameters as named by the developer. Reasoning before answering: none (one pass over the input); auto (the model decides for each request whether to think first, with at most 4,096 thinking tokens); medium effort (the provider's reasoning-effort setting); or not disclosed. Releases and settings are listed in Table 6.

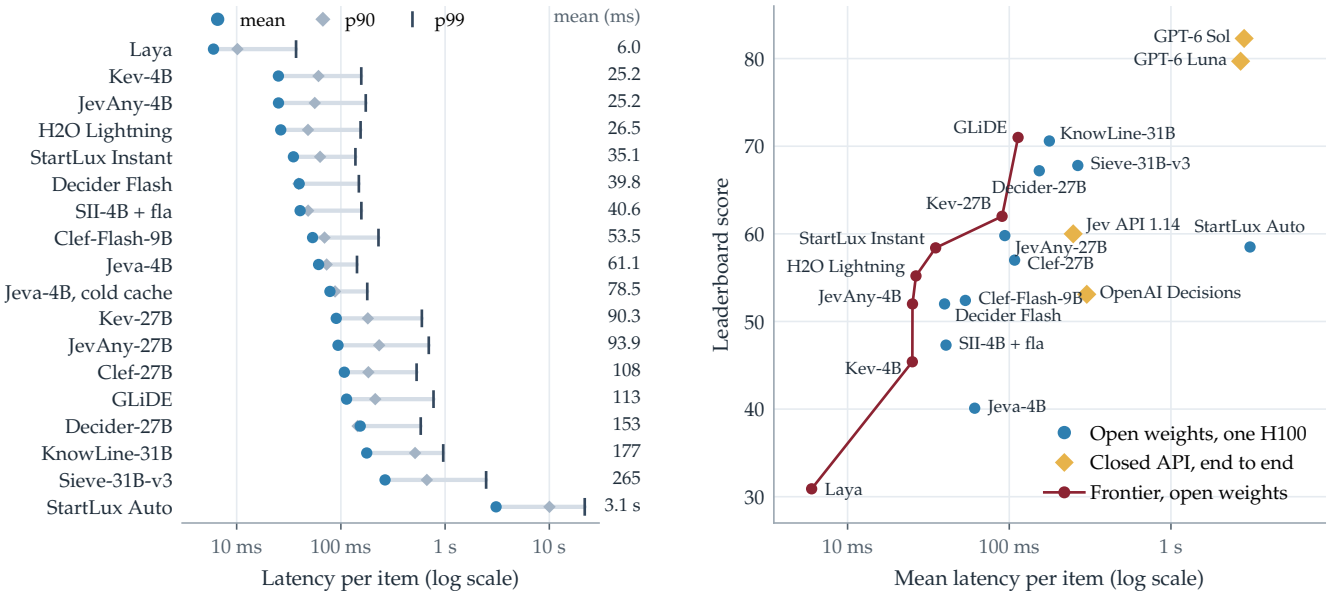
Configuration	Developer	Size	Access	Reasoning	Configuration	Developer	Size	Access	Reasoning
GLiDE [24]	Fastino	27B	open	none	Decider Flash [58]	Perplexity	4B	open	none
KnowLine-31B [56]	PelaAI	31B	open	none	JevAny-4B [76]	SimpleJev	4B	open	none
Sieve-31B-v3 [67]	Sthanika AI	31B	open	none	SII-4B [23]	SII	4B	open	none
Decider-27B [57]	Perplexity	27B	open	none	Kev-4B [54]	J. Palmer	4B	open	none
Kev-27B [53]	J. Palmer	27B	open	none	Jeva-4B [45]	Q. Li, X. Wang	4B	open	none
JevAny-27B [76]	SimpleJev	27B	open	none	Laya [16]	ConvAI Innovations	0.4B	open	none
StartLux Auto [86]	StartLux	4B	open	auto	GPT-6 Sol	OpenAI	–	API	medium effort
StartLux Instant [86]	StartLux	4B	open	none	GPT-6 Luna	OpenAI	–	API	medium effort
Clef-27B [13]	Cloudflare	27B	open	none	Jev API 1.14 [1]	TypeSafe	–	API	not disclosed
H2O Lightning [33]	H2O.ai	4B	open	none	OpenAI Decisions	OpenAI	–	API	not disclosed
Clef-Flash-9B [14]	Cloudflare	9B	open	none					

Table 3: **Scores by domain and difficulty tier.** E, M, H, and V denote the easy, medium, hard, and very hard tiers; All marks untiered domains. Rows follow the leaderboard score (Equation (5)). Darker is higher; bold is the best in each column. Tiers are not calibrated across domains.

		Sent.	Intent			Math				Code			Science			Know.	Safety			Retrieval			Verification			Tools			Web		
Model	**Score**	All	E	M	H	E	M	H	V	E	M	H	E	M	H	All	E	M	H	E	M	H	E	M	H	E	M	H	E	M	H
GPT-6 Sol	82.3	88	**100**	78	**82**	**100**	**100**	**100**	**100**	98	97	**93**	98	**96**	**91**	94	84	82	75	**99**	**91**	62	**100**	**99**	**100**	**99**	**88**	66	98	76	**53**
GPT-6 Luna	79.7	85	99	77	81	**100**	**100**	**100**	97	98	93	86	98	**96**	88	90	82	80	70	98	**91**	62	**100**	98	**100**	96	86	**68**	99	77	48
GLiDE	71.0	**90**	**100**	**84**	76	98	86	64	30	98	67	66	96	87	55	95	**87**	**84**	75	**99**	84	57	**100**	**99**	**100**	97	81	65	99	**81**	49
KnowLine-31B	70.6	88	**100**	83	78	98	98	71	40	**100**	74	71	98	94	52	93	86	83	77	**99**	89	**64**	97	96	90	97	84	56	**100**	74	44
Sieve-31B-v3	67.8	88	99	81	68	**100**	94	66	47	98	66	67	98	93	55	93	86	82	**78**	98	84	64	97	98	70	96	86	60	99	60	43
Decider-27B	67.2	89	98	83	70	96	90	61	33	**100**	70	67	**100**	90	52	95	86	81	75	**99**	88	56	99	97	65	97	84	67	99	79	50
Kev-27B	62.0	84	97	82	68	**100**	78	56	30	**100**	79	61	98	87	54	92	79	73	64	98	87	59	89	89	45	97	81	62	**100**	78	45
Jev API 1.14	60.0	86	99	78	56	**100**	84	63	30	95	76	54	96	94	71	92	83	73	69	**99**	83	60	88	88	20	97	75	59	99	73	41
JevAny-27B	59.8	84	98	81	52	**100**	82	53	27	**100**	73	66	**100**	87	60	94	81	75	65	92	70	57	93	90	35	98	84	58	99	71	41
StartLux Auto	58.5	84	96	71	70	98	90	81	47	98	70	53	90	86	51	85	74	66	62	98	82	56	88	76	35	92	76	46	92	50	31
StartLux Instant	58.4	88	94	71	70	98	88	43	40	98	71	50	96	84	46	88	82	73	63	**99**	82	54	86	80	55	96	79	54	98	39	31
Clef-27B	57.0	85	97	79	64	96	74	47	17	**100**	63	61	98	91	51	94	78	75	59	97	88	61	88	83	15	92	86	55	98	73	43
H2O Lightning	55.2	85	94	66	63	98	72	44	43	93	59	50	92	80	44	91	82	68	64	98	87	61	85	80	40	96	66	38	94	36	28
OpenAI Decisions	53.1	74	99	77	58	74	46	34	30	**100**	79	60	94	81	46	86	71	67	63	98	84	55	79	75	40	86	61	38	98	68	31
Clef-Flash-9B	52.4	83	99	42	64	94	68	51	17	93	69	50	98	94	48	**96**	70	66	46	**99**	83	58	76	58	35	91	50	32	90	55	25
Decider Flash	52.0	88	98	75	59	96	76	40	20	98	79	53	92	83	40	91	79	65	61	**99**	80	56	88	83	15	95	68	41	94	56	34
JevAny-4B	52.0	83	97	67	61	92	60	39	20	97	84	59	96	84	44	90	69	63	46	91	72	52	71	63	45	93	72	41	94	48	19
SII-4B	47.3	78	91	64	40	62	42	39	27	98	67	44	90	81	45	84	71	60	45	95	84	54	82	71	60	84	43	20	91	49	15
Kev-4B	45.4	74	92	64	50	90	58	33	20	**100**	**99**	49	88	74	41	82	66	58	32	93	80	54	76	72	50	81	46	22	83	39	16
Jeva-4B	40.1	73	83	48	51	56	36	30	23	**100**	74	53	70	50	29	65	48	48	37	95	74	58	73	54	30	51	27	9	84	20	10
Laya	30.9	65	82	7	30	42	42	23	20	53	49	51	26	29	24	43	49	39	35	86	62	38	52	44	30	1	1	0	44	6	2

Setup. We time the 17 open-weight configurations on the 1,109 single-question choice tasks of seven static domains, across all tiers. Each configuration runs the fast path documented by its authors on one NVIDIA H100 80 GB GPU, one request at a time. A task's latency runs from the request to the returned option probabilities and is the mean of three timed passes; we report the mean over tasks and the 90th and 99th percentiles. Appendix D gives the full protocol.

Latency. Figure 4a orders the configurations by mean latency. Apart from StartLux Auto, every configuration with at most 9B parameters averages 6.0 to 78.5 ms per task, with a 99th percentile of at most 230 ms. The 27B and 31B configurations average 90.3 to 265 ms, with 99th percentiles of 533 ms to 2.5 s. StartLux Auto, which reasons before answering on 408 of the 1,109 tasks, averages 3.1 s.



(a) Latency, open-weight configurations

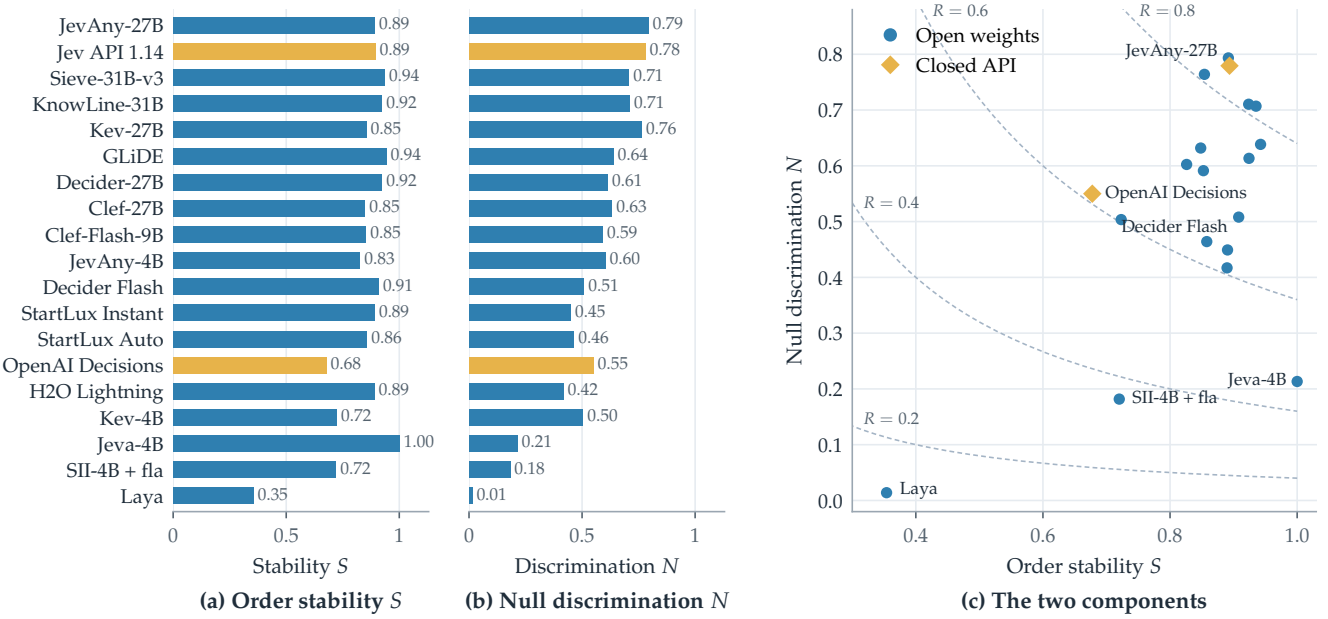
(b) Leaderboard score and latency

Figure 4: Without reasoning, open-weight System One configurations average 6.0 to 265 ms per task, and none slower than GLiDE scores higher. (a) Latency per task of the 17 open-weight configurations, with Jeva-4B timed with a warm and a cold option cache, on 1,109 single-question choice tasks, one request at a time on one H100 80 GB GPU: mean (dot; value at right), 90th percentile (diamond), and 99th percentile (bar). (b) Leaderboard score (Equation (5)) against mean latency. The line joins the open-weight configurations that no other open-weight configuration beats in both. Closed APIs (amber) are placed at their end-to-end times, which include the network and the provider's queue and are not comparable with the local times. SII-4B + fla adds flash-linear-attention to the documented setup of SII-4B (Appendix C).

Score and latency. Figure 4b places each configuration's leaderboard score against its mean latency. Among the open-weight configurations, the frontier rises from Laya (6.0 ms, 30.9) through four 4B configurations and Kev-27B (90.3 ms, 62.0) to GLiDE (113 ms, 71.0), and no slower open-weight configuration scores higher than GLiDE. Within a mean of 100 ms, the highest score is therefore Kev-27B's 62.0. StartLux Auto, which reasons before answering, takes 88 times the mean latency of StartLux Instant (3.1 s against 35.1 ms) and scores 58.5 against 58.4.

4.4. Reliability

The two reliability tests measure different properties: a configuration that keeps its answer when the options are reordered does not necessarily notice when the correct option is missing (Figure 5). We report both components of R (Equation (8)) for the 19 configurations with reliability runs: order stability S (Equation (6)) and null discrimination N (Equation (7)). Neither component follows the leaderboard: the highest S belongs to Jeva-4B, 20th of 21 by leaderboard score, and the highest N and R to JevAny-27B, ninth (Figure 1b). Across the 19 configurations, the two components are weakly related (Spearman $\rho = 0.37$; Figure 5c). Jeva-4B is fully stable ($S = 1.00$) but reaches only $N = 0.21$, and Decider Flash pairs $S = 0.91$ with $N = 0.51$. A high S is therefore no evidence that a model recognizes a missing answer, and the two properties must be measured separately.



(a) Order stability S **(b) Null discrimination N** **(c) The two components**

Figure 5: **Order stability and null discrimination measure different properties.** (a) Order stability S (Equation (6)) over 939 items, each in its original order and 12 shuffled orders; 1 is complete agreement and 0 the agreement expected by chance. (b) Null discrimination $N = 2\,\mathrm{AUC} - 1$ (Equation (7)) over 597 items, each with and without its correct option; 0 is chance. (c) N against S; dashed curves mark constant reliability $R = \sqrt{SN}$ (Equation (8)). Shown are the 19 configurations with reliability runs; rows in (a) and (b) follow R, and amber marks closed APIs.

4.4.1. Option Order

Sensitivity to option order, documented for LLMs that answer multiple-choice questions [59, 83], is low for most System One configurations: 15 of the 19 reach S between 0.83 and 1.00 (Figure 5a), and the open-weight 27B and 31B configurations reach 0.85 to 0.94. Four configurations fall below: OpenAI Decisions (0.68), SII-4B + fla (0.72), Kev-4B (0.72), and Laya (0.35), which changes its answer in 51% of reorderings. Jeva-4B reaches $S = 1.00$ by construction: it encodes each option separately, so the order of the options cannot change its scores.

4.4.2. Null Awareness

Null discrimination separates the configurations more widely, from $N = 0.01$ (Laya) to 0.79 (JevAny-27B; Figure 5b). The open-weight configurations with at most 9B parameters reach at most 0.60, and the 27B and 31B configurations reach 0.61 to 0.79. Scaling within a family raises N in all four families with two sizes: JevAny from 0.60 to 0.79, Decider from 0.51 to 0.61, Clef from 0.59 to 0.63, and Kev from 0.50 to 0.76. Of the closed APIs, Jev API 1.14 reaches 0.78 and OpenAI Decisions 0.55. Three configurations stay below 0.25: Jeva-4B (0.21), SII-4B + fla (0.18), and Laya (0.01).

5. Analysis

We analyze where System One models fall short of a reasoning model and what closes the gap (Section 5.1).

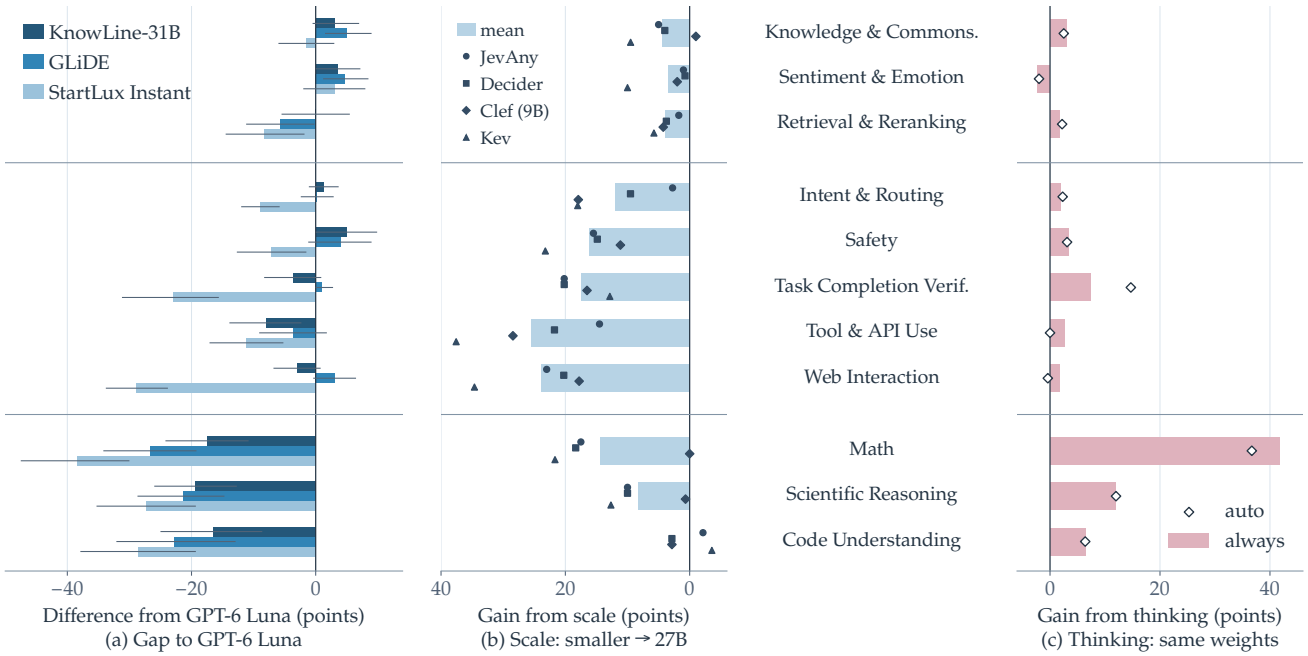


Figure 6: **System One models trail a reasoning model at every size in three domains, and scale and thinking help different domains.** Rows are the 11 domains, grouped by the patterns in (a). Scores are on the medium and hard tiers and the two untiered domains (2,061 tasks; Appendix E). (a) Score difference from GPT-6 Luna on the same tasks, with 95% bootstrap intervals over tasks. (b) Gain from the smaller to the 27B model of the same family: bars average the families, and markers show each family. (c) Gain from thinking for StartLux-4B-AutoThink with the same weights and a budget of 4,096 thinking tokens: *always* minus *never* (bars) and *auto* minus *never* (diamonds). Values are listed in Table 7.

5.1. Where System One Models Fall Short, and What Closes the Gap

GLiDE, the open-weight configuration with the highest leaderboard score, comes within 5.8 points of GPT-6 Luna, or exceeds it, in eight of the 11 domains, and the 4B StartLux Instant in two. In Math, Scientific Reasoning, and Code Understanding, System One models trail at every size we evaluate, from 4B to 31B. Scaling a model and letting it reason before answering help different domains (Figure 6).

The gap to a reasoning model. Figure 6a gives the score difference from GPT-6 Luna on the same tasks for three System One models: GLiDE (27B); KnowLine-31B, second by leaderboard score among open-weight configurations; and StartLux Instant, the highest-scoring 4B configuration that does not reason before answering. The domains fall into three patterns. In Knowledge & Commonsense and Sentiment & Emotion Understanding, all three are within 1.5 points of Luna or ahead of it, the 4B model included. In Intent & Routing, Safety, Task Completion Verification, Tool & API Use, and Web Interaction, the gap depends on size: StartLux Instant trails Luna by 7.1 to 28.8 points, while GLiDE lies between -3.6 and $+4.0$. In Math, Scientific Reasoning, and Code Understanding, all three trail, by 16.4 to 19.3 points for KnowLine-31B and by 27.3 to 38.3 points for StartLux Instant. Retrieval & Reranking fits none of the three patterns: the three models trail Luna by 0.0 to 8.2 points. The grouping is descriptive: we formed it after inspecting the results.

Scale and thinking. The two kinds of gap suggest two ways to raise a System One model's score: a larger model, and reasoning before answering. We measure each with the other held fixed. For scale, we take the score difference between the 27B model and the smaller model of each of four families released in both sizes (JevAny, Decider, Clef with 9B, and Kev) and average it over all four families in every domain (Figure 6b). For thinking, we run StartLux-4B-AutoThink with the same weights in its three thinking modes, *never*, *auto*, and *always*, with a budget of 4,096 thinking tokens, on all 2,061 tasks, and score each mode with the benchmark's per-task credit (Figure 6c; details in Appendix E).

The two help different domains. Thinking raises StartLux-4B-AutoThink by 41.7 points in Math and by 12.0 in Scientific Reasoning, more than scale does there (14.4 and 8.3). In the five domains where the gap depends on size, scale raises scores by 12.0 to 25.6 points, and every family gains in each of them, by 2.8 (JevAny, Intent & Routing) to 37.6 points (Kev, Tool & API Use); thinking in the *always* mode adds 1.8 to 7.3 points. Knowledge & Commonsense, Sentiment & Emotion Understanding, and Retrieval & Reranking gain at most 4.4 points from either. Code Understanding gains too little from either to close its gap: 0.0 points from scale and 6.4 from thinking, against a gap of 22.9 points between GLiDE and Luna. That thinking helps most in Math is consistent with the finding that chain-of-thought helps mainly on math and symbolic reasoning [66]. Test-time computation has been compared with model size on mathematics [64]; our comparison covers 11 domains. These comparisons rest on one thinking model at one budget and on four families, and we do not test why scale helps where thinking does not.

Implication. The domain determines what closes the gap to a reasoning model. In Knowledge & Commonsense and Sentiment & Emotion Understanding, the 4B StartLux Instant is within 1.5 points of GPT-6 Luna or ahead; in the five size-dependent domains, the 27B GLiDE is within 3.6 points or ahead. In Math, thinking raises StartLux-4B-AutoThink from 50.8 to 92.5, against Luna's 100.0. In Scientific Reasoning and Code Understanding, neither KnowLine-31B nor StartLux-4B-AutoThink with thinking comes within 16 points of Luna.

6. Related Work

6.1. System One Models and Decision Benchmarks

System One models expose predictions over explicitly specified outcomes, enabling classification, routing, scoring, and verification without open-ended text generation. Jev introduced this interface as a general-purpose service [1], while open releases such as Kev and JevAny provide language-model-based implementations [54, 76]. Recent research investigates how these capabilities can be obtained from pretrained models: SII-Decision distills reasoning-informed predictions into a single-pass readout, whereas Releasing the Inner Jev studies answer-token readout and supervised adaptation [23, 28]. Evaluation efforts already extend beyond accuracy. JevBench measures capability, calibration, latency, and cost [4]; JEVal covers ten application domains and examines how local decisions compose into agent workflows and social simulations [23]. OpenDecisions additionally includes difficulty tiers and controlled perturbations [28]. Our work complements these efforts with within-domain difficulty tiers, controlled history-dependent verification, and a separate reliability score within a broader evaluation of decision competence.

6.2. Generalist Capability Evaluation

MMLU measures knowledge and problem solving across academic and professional subjects [34], while BIG-Bench Hard probes challenging tasks and the effects of explicit reasoning [69]. Humanity's Last Exam extends evaluation toward expert-level academic questions [8]. These benchmarks establish useful measures of capability breadth and demanding problem solving. HELM further emphasizes standardized evaluation across scenarios and multiple desiderata, including calibration and efficiency [47]. System One's Last Exam adopts a complementary focus: assessing generalist competence through decisions over supplied evidence and specified alternatives, while also examining whether judgments remain reliable as their informational and temporal requirements change. Academic reasoning is therefore one component of our evaluation, alongside semantic, policy, evidence, and environment-dependent decisions.

6.3. Judges, Reward Models, and Verification Reliability

Research on LLM-as-a-judge establishes the usefulness of automated evaluation while documenting position, verbosity, and self-enhancement biases [84]. RewardBench evaluates preferences across chat, safety, and reasoning, whereas JudgeBench constructs response comparisons grounded in factual and logical correctness [41, 71]. VerifyBench examines agreement between candidate responses and reference answers across scientific domains, including sensitivity to response presentation [46]. These studies motivate scrutinizing the evaluator itself. Separately, calibration research distinguishes predictive accuracy from the quality of confidence estimates, and selective classification studies the trade-off between error and coverage [26, 32]. We connect these perspectives to general-purpose decision evaluation, encompassing judgments whose evidence consists of documents, structured states, or interaction histories rather than only candidate answer pairs.

6.4. Evaluation in Agentic Environments

Mind2Web evaluates action prediction across tasks, websites, and domains [22]. WebArena provides reproducible web environments and functional evaluation using answers and programmatic state checks [85]; τ-bench evaluates policy-constrained tool interaction using database outcomes and repeated-trial reliability [79]. More recently, VerifierBench evaluates the generation of executable checkers against valid solutions, ordinary errors, and adversarial cases [80]. These works address complementary evaluation targets: executing tasks, maintaining reliable behavior, or constructing verification programs. We evaluate decision models in two of these roles: as the policy that chooses each action of a tool or web episode from a menu of public options, and as the verifier that judges recorded executions, including trajectories whose final goal projections agree while their histories differ (Appendix A.4).

7. Conclusion

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A. Benchmark Construction Details

This appendix complements Section 3 with per-domain construction details. Every item records its source, pinned version, original split, and original identifier; these records, together with the reference of each item, are kept separate from the model inputs.

A.1. Semantic Understanding

Sentiment & Emotion Understanding (200). Amazon Reviews 2023 [37] contributes 50 `score` items in which the model predicts the reviewer's recorded 1–5 star rating from the review title and text (10 per rating, drawn from the All Beauty category). GoEmotions [21] contributes 40 single-label test comments, four for each of ten target emotions, each answered over the full 28-label set. SST-5 [65] contributes 30 sentences (six per class), DynaSent [60] 50 test sentences (18 negative, 18 positive, 14 neutral), and ARTS [78] 30 aspect-level restaurant items derived from 10 source sentences. Classification options are shuffled. A recorded star rating is not always recoverable from the text, and the emotion labels measure agreement with the source annotation rather than an uncontested semantic truth.

Intent & Routing (300). The easy tier contains 100 single requests from CLINC150 [42] (42), BANKING77 [6] (29 in-scope, 11 out-of-scope against a 50-category service catalog), and SGD [62] (18), posed either as a four-way `choice` between the native intent and three neighboring native classes (47) or as a `noul` question about one candidate intent (53). The medium tier contains 100 complete NatCS [31] customer-service conversations (insurance 36, banking 30, finance 34) with exactly two customer goals, which are often far apart in the dialogue; the model answers one `noul` question per native goal label (22–39 per conversation) and must ignore goals suggested only by the agent. The hard tier contains 100 NLU++ [7] utterances (banking 73, hotels 27), each with one `noul` question per ontology component (48 or 40) using the native definitions verbatim. Medium and hard items are scored by the per-item Jaccard index, where the predicted set contains the labels whose affirmative probability exceeds one half. Medium and hard items were drawn by deterministic hash ranking and then screened by source-only audits, which held out 30 and 20 candidates, respectively, for label uncertainty or task fit.

A.2. Knowledge and Reasoning

Math (200). The easy tier combines GSM8K-Platinum [15, 74] problems with at most three calculation steps (30) and MATH500 [35, 48] Level 1 problems (20); the medium tier combines GSM8K-Platinum problems with four or more steps (20) and MATH500 Levels 2–3 (30); the hard tier contains MATH500 Levels 4–5 (70); and the very hard tier contains AIME 2025 (13) and AIME 2026 (17) problems [3]. MATH500 is restricted to problems with numeric answers and without figures. Of the 600 distractors, 350 are wrong answers sampled from Qwen3.5-4B-Base without reasoning (16 samples per problem), 22 are intermediate results of GSM8K solutions, and 228 are rule-based near misses. The options are sorted by value, and the position of the correct answer is balanced. A candidate is dropped if its correct answer can be identified from the options alone or if its label appears doubtful.

Code Understanding (200). Each item presents a LeetCode problem statement and a Python solution and asks whether the solution returns the correct answer for every valid input. CodeJudgeBench [39] contributes 156 solutions written by seven LLMs; 44 further solutions were sampled from Qwen3.5-4B-Base for 2024–2025 LeetCodeDataset [77] easy problems that are not in CodeJudgeBench. Comments

and docstrings are removed. Every solution is re-executed in a network-isolated sandbox on the public and private LiveCodeBench [38] tests under the original import header. An incorrect solution must fail by a wrong answer or a runtime error; solutions that fail only by time, memory, or recursion limits are dropped because such failures depend on the machine. A correct solution must also pass LeetCodeDataset's tests of the same problem or a recorded manual check, and every correct solution in the release was compared with an independent brute-force solution on thousands of valid random inputs. Each incorrect solution fails one of the statement's own examples or a test whose expected output an independent solution confirms. An integer program selects 60/70/70 items by judging difficulty. Easy items are restricted to LeetCode Easy or Medium problems whose incorrect solutions fail more than 20% of the tests, and hard items to Medium or Hard problems whose incorrect solutions pass at least 30% of the tests, so that their errors surface only on some inputs. The selection keeps one item per problem family (a problem and its variants), half of the solutions correct in each tier, and correct and incorrect solutions matched in LeetCode level, generator, and code length.

Scientific Reasoning (200). The easy tier contains 50 ARC-Challenge [12] test questions with four options, the medium tier 70 MMLU-Redux [27] questions from natural-science subjects that its re-annotation marks as error-free, and the hard tier 80 GPQA Diamond [63] questions (36 chemistry, 35 physics, 9 biology), whose options are shuffled with a fixed seed.

Knowledge & Commonsense (200). The domain contains 120 error-free MMLU-Redux questions from 36 non-STEM subjects (at most six per subject), 40 HellaSwag [82] validation items, and 40 CommonsenseQA [70] validation items, from which one randomly chosen distractor is removed to leave four options. Mathematics, computer-science, and logic subjects of MMLU are used in neither domain, and natural-science subjects appear only in Scientific Reasoning. Math, Science, and Knowledge exclude items with duplicate option texts, Science and Knowledge also exclude items found wrong or ambiguous in an earlier internal label audit, and the gold position is balanced within each source.

A.3. Safety and Policy Compliance

Safety (500). RAGTruth [51] contributes 150 items: 90 ask whether a response contains a hallucination with respect to the supplied material, and 60 ask which segment, if any, contains the only hallucinated span. ReasoningShield [43] contributes 75 reasoning traces with human majority labels (45 harmful or safe, 30 three-level risk). HarmThoughts [40] contributes 50 items that ask for the behavior category of a target step among 16 categories. DynaBench [36] contributes 100 items that ask whether a dialogue complies with a given policy (50) or which rule, if any, it violates (50). Aegis 2.0 [29] contributes 50 prompt-safety and 50 response-safety items, and OR-Bench [19] contributes 25 hard benign prompts that test over-refusal. One hundred references are model-derived: the original LLM annotations of HarmThoughts, the LLM-jury labels of 25 Aegis responses, and the OR-Bench hard-benign membership. Tiers (177/165/158) are content-based grades of the reasoning an item requires, assigned during curation without model predictions or independent calibration; for example, they separate a direct refusal from a disguised context, or a directly matching rule from multi-hop references and interacting exceptions.

A.4. Evidence Assessment

Retrieval & Reranking (300). All items are pointwise relevance judgments. The easy tier contains 100 TREC Deep Learning 2019/2020 passage pairs [17, 18] restricted to the endpoint judgments (perfectly relevant versus irrelevant; 50 each). The medium tier combines 70 BRIGHT [68] pairs from biology, economics, sustainable living, and earth science (35 relevant), whose negatives are non-gold corpus passages retrieved with BM25 and screened with LLM assistance, with 30 Touché 2020 pairs [5, 72] (ten topics, one pair per native grade 0–2). The hard tier contains 100 ACORD [75] requirement–clause pairs, 25 for each native grade 1–4 on a 0–4 scale, which receive adjacent-grade credit. Pairs of the same query are not independent.

Task Completion Verification (200). BrowseComp-Plus [10] contributes 50 items that ask whether the snippets retrieved by a fixed historical search run establish a newly defined subgoal; each of 25 search prefixes yields one fully and one partially supported item, and their references are new model-derived annotations. CALVIN [50] contributes 8 light-state checks on official debug states, τ-bench [79] 56 checks of backend records after replaying historical tool calls, and MiniGrid [11] 46 state-predicate checks on newly collected seeded rollouts. The Embodied Agent Interface [44] contributes 40 VirtualHome items: 10 easy and 10 medium current-state checks, and 20 hard history-dependent items. Tiers follow the information and operations a decision requires, measured by its minimum sufficient solution: a direct check of the current state, bounded integration of records or evidence, or verification over the execution history. Each hard item extends the successful original action prefix of one of five base tasks with an executable audit suffix that switches an appliance on or off, removes two target objects, and restores them fully, partially, or not at all. C1 holds if the suffix switches an appliance on at least once and all required object placements hold before every such action; C2 holds if the suffix contains a close action and all required placements hold before the first one; and C3 holds if all original state and placement goals hold after the last action. Switch-on and close actions of the original prefix are outside the scope of C1 and C2, and a later restoration does not remove an earlier violation. The four options are the correct three-bit profile and the three profiles that differ from it in one bit, so every condition must be judged. C3 checks the final state, whereas C1 and C2 check the execution history: for every item, an executable trajectory reaches the same final goal state but has the opposite value of C1 or C2. For each base task, all eight profiles were generated and verified; four enter the benchmark, and the others serve as counterfactual witnesses. Verification has three layers: the native executor accepts the prefix and the suffix action by action; an independent state machine reconstructs every projected state from the item text; and goal atoms are checked against the upstream oracle before the three conditions are computed. Each condition is true in 10 of the 20 items. A negative answer means that the visible evidence does not establish the requirement, not that the requirement is false. Because the 20 hard items come from five base tasks, they should be resampled by base task when estimating uncertainty.

A.5. Agent Task Completion

Tool & API Use (282). The easy tier contains 100 BFCL [55] multi-turn requests, 26 at the first turn and 74 at later turns whose earlier turns are replayed as public history, each asking for an explicitly specified operation or a direct lookup. The medium tier contains 78 ToolSandbox [49] scenarios reduced to one self-contained request, including continuations whose earlier turns are executed in the simulator and shown as history. The hard tier contains 104 original first-turn BFCL multi-turn requests that involve dependent lookups, conditional actions, or several coordinated outcomes; these features describe the tier as a whole,

and individual tasks vary. BFCL tasks are scored 0/1 by the unchanged BFCL checker. ToolSandbox tasks use the native milestone score with two adaptations: the wording checks of final acknowledgements are dropped, since the model cannot write prose, and a task scores 1 whenever the full goal state is reached without a minefield violation. Episodes have budgets on model decisions and native tool calls (for example, 96 decisions and 12 calls for hard BFCL tasks), and exhausting a budget is a scored outcome. A request with a single option is answered automatically, and tool execution is bounded in time, memory, and output size; a call that exceeds a bound returns an error and has no effect.

Text-based Web Interaction (337). The easy tier contains 100 single steps of Mind2Web [22] demonstrations: 50 ask whether a supplied action advances the goal, and 50 ask for the best of four supplied actions, whose distractors are other controls on the same page. The operation and any typed value are supplied, so this tier tests element grounding. The medium and hard tiers contain 137 and 100 BreakingWeb [81] tasks in seven simulated applications, with 361 and 969 scored decisions on the target entity, the exact value to enter, and when to stop. A hard task has at least five scored decisions or at least three entity choices not named in the instruction, and at least one hard decision: a value recalled from an earlier page, a menu with more than 60 options, or several inferred entities. Filling a field is decomposed into choosing the field, a group of candidate values when there are more than 64, and the exact text span. Reference routes were found by an external LLM or written by hand, and each was replayed to success under the unchanged checker. Alternative orders of the route and detours that do not prevent success are verified by replay and accepted.

A.6. Reference Corrections and Usage Restrictions

Offline LLM assistance in curation, such as route finding for web tasks and the review of retrieval negatives, never takes part at evaluation time.

Reference corrections. After evaluation, a per-item review corrected 18 references whose source labels contradicted the item's own material or stated rules: 17 in Safety (7 Aegis 2.0, 6 RAGTruth, 2 ReasoningShield, and 2 HarmThoughts) and one BrowseComp-Plus item of Task Completion Verification. The review was a single LLM-assisted adjudication pass, not multi-rater agreement. Inputs, options, membership, and scoring rules were unchanged; the original labels are kept as provenance, and every model was rescored from its unchanged predictions. Seven corrected items belong to the leaderboard selection, all in the hard tier of Safety.

The 150 model-derived references (100 in Safety and the 50 BrowseComp-Plus items of Task Completion Verification) are included in all reported scores.

Usage restrictions. The complete benchmark is intended for research use, and several sources restrict redistribution: Mind2Web asks that its test data not be published online, BreakingWeb ships without a license, GPQA asks that its questions not be revealed in plain text online, and AIME 2026 (CC BY-NC-SA 4.0) and ReasoningShield (CC BY-NC 4.0) prohibit commercial use. All other sources retain their original terms.

B. Refusals and Over-Window Inputs

Every configuration must return a decision for every request it receives. An explicit refusal, in which the API returns a refusal object instead of a distribution over the options, is scored like any other failure to decide: the static task or tool episode receives zero credit, and a refused web decision counts as wrong, so route progress stops there. Inputs are never truncated or compressed to fit a model; a task with any request longer than the model's native context window receives zero credit. Across the 21 configurations, these rules affect only Laya (Table 4) and OpenAI Decisions (Table 5); every other configuration returned a valid decision for every request.

Over-window inputs. Laya reads at most 8,192 tokens. In 30 web tasks (9 medium, 21 hard), at least one reference-route request exceeds this window; together they contain 137 over-window requests. Each such task receives zero credit, including 11 decisions answered correctly before the first over-window request. Under progress scoring, 29 of the 30 tasks would score zero even without this rule, because an earlier in-window decision was already wrong; the remaining task, jev-2786, exceeds the window at its first request. The 21 hard tasks belong to the leaderboard selection.

Table 4: **Web tasks that exceed Laya's 8,192-token window.** "Requests" counts the model requests of the reference route; "First over" gives the first request beyond the window and its length in tokens. Every task receives zero credit.

Task	Tier	BreakingWeb task	Requests	First over (tokens)
jev-2785	M	`reddit_account_cleanup`	3	3 (8,689)
jev-2786	M	`reddit_block_and_cleanup`	1	1 (9,137)
jev-2787	M	`reddit_comment_save_settings`	5	5 (8,822)
jev-2789	M	`reddit_discover_subscribe_post`	7	7 (8,571)
jev-2790	M	`reddit_edit_delete_subscribe`	7	7 (9,278)
jev-2793	M	`reddit_manage_subscriptions`	6	6 (8,475)
jev-2794	M	`reddit_multi_sub_engagement`	5	5 (9,246)
jev-2796	M	`reddit_notification_driven_workflow`	3	3 (9,110)
jev-2803	M	`reddit_selective_vote_save`	5	5 (8,804)
jev-2820	H	`amazon_budget_optimized_spree`	6	4 (8,671)
jev-2830	H	`booking_expert_compare_and_decide`	23	10 (8,551)
jev-2831	H	`booking_expert_complete_account_review`	36	32 (12,837)
jev-2832	H	`booking_expert_family_vacation`	17	13 (8,535)
jev-2834	H	`booking_expert_multi_city_booking`	29	11 (9,108)
jev-2835	H	`booking_expert_notification_workflow`	23	19 (8,264)
jev-2836	H	`booking_expert_rebooking_workflow`	15	7 (8,315)
jev-2838	H	`booking_expert_travel_planner`	16	12 (10,907)
jev-2839	H	`booking_frontier_cancel_and_reorganize`	14	5 (8,417)
jev-2840	H	`booking_frontier_grand_tour`	27	13 (8,958)
jev-2868	H	`gmail_vacation_preparation`	26	24 (8,991)
jev-2907	H	`reddit_complete_account_setup`	13	9 (10,393)
jev-2908	H	`reddit_complete_engagement_cycle`	13	5 (8,989)
jev-2909	H	`reddit_content_curation`	11	11 (8,487)
jev-2910	H	`reddit_cross_platform_workflow`	19	19 (9,400)
jev-2912	H	`reddit_mass_engagement_workflow`	23	19 (8,693)
jev-2913	H	`reddit_multi_community_outreach`	26	14 (8,640)
jev-2915	H	`reddit_research_and_create`	25	21 (8,305)
jev-2916	H	`reddit_research_respond`	15	15 (8,930)
jev-2917	H	`reddit_search_and_engage`	16	16 (8,266)
jev-2918	H	`rh_cross_reference_1099`	11	11 (8,275)

Refusals. OpenAI Decisions refuses 22 requests. The three static Safety refusals are HarmThoughts items that ask only for the functional category of one step in a harmful reasoning trace, and the Intent & Routing refusal concerns one of 48 binary label questions about a benign banking message, which voids the whole label set. Every refusal is a bare refusal object with no explanation and zero output tokens, `{"type": "refusal", "name": "q"}`. Most refused tool requests are benign, such as booking a flight or starting a car engine. Five tool refusals occur at the choice of the next function. The other seven occur at an argument value, in four cases a credential field (`client_id` or `client_secret`) after 19 to 26 earlier decisions; in all seven, the model had already chosen a function whose required value was not yet available, so the menu offered only unrelated grounded strings, such as `TradingBot` and `trading_get_login_status` as the stock symbol of an Apple order. All six web refusals occur at exact-value steps. Seventeen of the 22 affected tasks belong to the leaderboard selection.

Table 5: **Refusals by OpenAI Decisions.** "Refused at" gives the model decision at which the refusal occurred. Credit is the task credit after the zero-credit rule. *In the leaderboard selection.

Task	Domain, tier	Request	Refused at	Credit
jev-0427	Safety, M	HarmThoughts trace about making a child adopt a parent's personality	sole question	0
jev-0435	Safety, E	HarmThoughts trace about how hard to spank children	sole question	0
jev-0444	Safety, E	HarmThoughts trace about tying a child to a chair	sole question	0
jev-1077*	Intent, H	NLU++: "I spent 67.27 euros from my current account this morning. What's my current balance?"	label 22 of 48	0
jev-2430	Tools, M	ToolSandbox: "Make all of my friends my enemy"	decision 1	0
jev-2444	Tools, M	ToolSandbox: "Turn all my friends in my contact book my enemy"	decision 1	0
jev-2472*	Tools, H	BFCL vehicle: start the engine after locking the doors and applying the parking brake	decision 1	0
jev-2480*	Tools, H	BFCL: estimate the distance from Rivermist to Stonebrook and text it to a cousin	decision 2	0
jev-2497*	Tools, H	BFCL vehicle: lock the doors, engage the parking brake, start the engine, and check tire pressure	decision 9	0
jev-2512*	Tools, H	BFCL trading: buy 100 Apple shares after confirming the market is open	decision 4	0
jev-2532*	Tools, H	BFCL travel: book a first-class flight from San Francisco to Los Angeles	decision 25	0
jev-2539*	Tools, H	BFCL travel: book a flight from Rivermist to Los Angeles	decision 20	0
jev-2540*	Tools, H	BFCL travel: book a flight from San Francisco to Los Angeles	decision 20	0
jev-2544*	Tools, H	BFCL travel: add travel insurance to the booked trip flight_001	decision 1	0
jev-2551*	Tools, H	BFCL travel: book a flight from Oakendale to Los Angeles	decision 8	0
jev-2553*	Tools, H	BFCL travel: check the cost of a business-class flight from Rivermist to Los Angeles, then book it	decision 27	0
jev-2873*	Web, H	BreakingWeb `lms_complete_account_audit`	one exact-value step	0
jev-2878*	Web, H	BreakingWeb `lms_ta_dual_role`	one exact-value step	0
jev-2879*	Web, H	BreakingWeb `lms_waitlist_strategy`	one exact-value step	0
jev-2906*	Web, H	BreakingWeb `pp_wellness_visit_prep`	one exact-value step	0.18
jev-2913*	Web, H	BreakingWeb `reddit_multi_community_outreach`	one exact-value step	0
jev-2916*	Web, H	BreakingWeb `reddit_research_respond`	one exact-value step	0.25

C. Configuration Details

Table 6 lists the release of each open-weight configuration with the settings of our latency and reliability runs, and the model identifier and settings of each API.

Exceptions to the documented settings. Laya's official package cuts each option text at 48 tokens; all runs use the package without this cut, so that every option is read in full. In our latency and reliability runs, SII-4B + fla adds flash-linear-attention to the documented setup of SII-4B; its leaderboard score is that of the documented setup.

Reliability runs. Each open-weight configuration ran in the settings of the latency measurements (Appendix D), one request at a time per model copy, on NVIDIA H100 80 GB GPUs; the two System One APIs ran with four requests in flight. Eight of the 18 corrected references (Appendix A.6) belong to the reliability sets. In the option-order test (six of the eight), we rescored the existing predictions; in the null-awareness test (all eight), we reran the affected items with the corrected option removed.

Scored runs and our runs. The scores of Section 4.2 come from runs separate from those behind the latency and reliability results. On the 939 option-order items in their original order, the two give the same credit on 907 to 938 items for each open-weight configuration except StartLux Auto, which samples its thoughts and agrees on 875. For SII-4B, the comparison is between its documented setup (scored) and SII-4B + fla (ours).

Table 6: **Releases and settings.** Open-weight configurations: the Hugging Face repository at the revision used, and the inference path of our latency and reliability runs, which follows each developer's documented fast path. APIs: the model identifier and settings of the evaluation runs.

Configuration	Release	Settings
GLiDE	`fastino/GLiDE` @ d199b6d	In process (`Glide.decide`), BF16, SDPA attention, flash-linear-attention
KnowLine-31B	`PelaAI/KnowLine-31B-Gen1` @ 8f052a8	SGLang 0.5.21, BF16 with an FP32 output layer, the documented read-twice front end
Sieve-31B-v3	`sthanika-ai/Sieve-31B-v3` @ 44aa517	In process (`load_sieve`), BF16, eager; the released policy, which scores each request in two option orders in one pass
Decider-27B	`perplexity-ai/pplx-decider-v1.2` @ cce0f3f	Documented fast path (layer compilation, CUDA graphs, length buckets) with FP8 matrix multiplication
Kev-27B	`jaredpalmer/kev-27b` @ af0e6d5	Documented serving defaults in process: BF16, fused kernels, CUDA graphs
JevAny-27B	`SimpleJev/JevAny-Qwen3.8-27B-LoRA` @ 09c9e91 on `Qwen/Qwen3.8-27B` @ 1d4bf0f	Adapter merged in BF16, SDPA attention, flash-linear-attention, CUDA graphs
StartLux Auto	`startlux-models/StartLux-Decision-v1.1-4B-AutoThink` @ 23dd4c5	Documented server on vLLM 0.19.1 with FP8; auto thinking with gate 0.6 and at most 4,096 thinking tokens
StartLux Instant	`startlux-models/StartLux-Decision-v1.1-4B-Instant` @ 416088f	Documented server on vLLM 0.19.1 with FP8
Clef-27B	`Cloudflare/clef` @ a30162b	Documented SGLang build (`dev-clef`)
H2O Lightning	`h2oai/h2o-lightning-4b` @ 08e1da5	The developer's vLLM 0.30.0 build with FP8
Clef-Flash-9B	`Cloudflare/clef-flash` @ f0b7d3c	As Clef-27B
Decider Flash	`perplexity-ai/pplx-decider-flash-v1.2` @ 9ff7cf9	Documented fast path, BF16
JevAny-4B	`SimpleJev/JevAny-Qwen3.5-4B-LoRA` @ 1c7aa9b on `Qwen/Qwen3.5-4B` @ 851bf6e	As JevAny-27B
SII-4B	`jylin001206/SII-Decision-4B-LoRA` @ 5c77219	`DecisionEngine`, BF16; flash-linear-attention added in our runs (SII-4B + fla)
Kev-4B	`jaredpalmer/kev-4b` @ 6cfce5c on `Qwen/Qwen3.5-4B-Base` @ 1001bb4	As Kev-27B, with the adapter merged
Jeva-4B	`LIQIIIII/Jeva-4B` @ 25ca454	`predict()` with the documented defaults; option cache warm or cold (Appendix D)
Laya	`convaiinnovations/laya` @ 7b928d8	TileLang fast path, 8,192-token input window; options not cut at 48 tokens
GPT-6 Sol	`gpt-6-sol`	Responses API, medium reasoning effort, at most 25,000 output tokens
GPT-6 Luna	`gpt-6-luna`	As GPT-6 Sol
Jev API 1.14	`jev-1.14.0`	TypeSafe decision API
OpenAI Decisions	`gpt-6-luna`	OpenAI decision API

D. Latency Measurement

Tasks. The 1,109 timed tasks are every single-question choice task of seven static domains, across all tiers: Math 200, Scientific Reasoning 200, Knowledge & Commonsense 200, Safety 190, Sentiment & Emotion Understanding 150, Task Completion Verification 122, and Intent & Routing 47.

Configurations. Each configuration runs the fast path documented by its authors (Table 6), at their settings for precision, fused kernels, and CUDA graphs, and uses FP8 only where its documentation specifies it. Jeva-4B caches option encodings across requests; we time it with the cache kept within a pass (warm) and with the cache cleared before every request (cold).

Timing. Each configuration runs on one NVIDIA H100 80 GB GPU used by no other process, with one request per call. The clock starts when the request is handed to the model's documented entry point and stops when the option probabilities are back on the host; it therefore includes tokenization and, for server-based models, the local HTTP round trip. After one untimed warm-up pass over all tasks, each configuration makes three timed passes, and a task's latency is the mean of its three timings. Two copies of each configuration ran at the same time on two GPUs, each on its own half of the tasks and pinned to its own 24 CPU cores.

Closed APIs. The closed APIs cannot be timed on this hardware. Figure 4b places them at their end-to-end times from the evaluation runs, which include the network and the provider's queue, with four requests in flight.

E. Analysis Details

Task selection. The analysis of Section 5.1 uses the medium and hard tiers of each tiered domain (for Math, its 50 medium and 70 hard tasks) and all tasks of the two untiered domains: 2,061 tasks, with 109 to 323 per domain (Table 7). The leaderboard selection (Equation (5)) has as few as 20 tasks in one domain (Task Completion Verification).

Per-domain values. Table 7 lists the values plotted in Figure 6, and Figure 7 places each domain's gain from scale against its gain from thinking.

Thinking control. StartLux-4B-AutoThink ran in each of its three thinking modes on all 2,061 tasks: the 1,642 static tasks (8,167 questions per mode), sent as batched requests, and the 419 Tool & API Use and Web Interaction tasks through the benchmark's runtime (5,500 to 5,800 model calls per mode). Our *auto* run gives the same credit as the benchmark's own *auto* run on 1,510 of the 1,642 static tasks (92%) and on 376 of the 419 tool and web tasks (90%); thinking is sampled, and our requests were batched.

Hard tiers only. Table 8 repeats the scale and thinking gains on the hard tier of each tiered domain, together with all tasks of the two untiered domains (1,202 tasks, with Math's 70 hard tasks). The patterns of Section 5.1 are unchanged except in three domains. In Intent & Routing, the scale gain falls from 12.0

to 4.9 points. In Task Completion Verification, whose hard tier has 20 tasks, the scale gain falls from 17.4 to 3.8 points and the thinking gain from 7.3 to −15.0. In Code Understanding, the scale gain rises from 0.0 to 11.4 points and the thinking gain falls from 6.4 to 1.4.

Table 7: **Per-domain values behind Figure 6.** Tasks: the medium and hard tiers, or all tasks of an untiered domain. Luna: GPT-6 Luna's score. Differences from Luna with 95% bootstrap intervals over tasks (4,000 resamples). Scale gain: the mean gain from the smaller to the 27B model of the same family. StartLux-4B-AutoThink: its scores in the three thinking modes, with the same weights.

			Difference from GPT-6 Luna			Scale	StartLux-4B-AutoThink		
Domain	Tasks	Luna	KnowLine-31B	GLiDE	StartLux Instant	gain	Never	Always	Auto
Knowledge & Commons.	200	90.0	+3.0 [−0.5, +7.0]	+5.0 [+1.5, +9.0]	−1.5 [−6.0, +3.0]	+4.4	85.0	88.0	87.5
Sentiment & Emotion	200	84.8	+3.5 [+0.0, +7.2]	+4.8 [+1.2, +8.5]	+3.0 [−2.0, +8.0]	+3.4	88.0	85.8	86.0
Retrieval & Reranking	200	76.5	+0.0 [−5.5, +5.5]	−5.8 [−11.2, −0.2]	−8.2 [−14.5, −1.8]	+3.9	69.0	70.8	71.2
Intent & Routing	200	79.5	+1.3 [−1.1, +3.7]	+0.1 [−2.4, +2.9]	−8.9 [−12.0, −5.8]	+12.0	68.8	70.8	71.1
Safety	323	75.2	+5.0 [+0.0, +9.9]	+4.0 [−1.2, +9.0]	−7.1 [−12.7, −1.5]	+16.2	60.4	63.8	63.5
Task Completion Verif.	109	98.2	−3.7 [−8.3, +0.9]	+0.9 [+0.0, +2.8]	−22.9 [−31.2, −15.6]	+17.4	58.7	66.1	73.4
Tool & API Use	182	75.8	−8.0 [−13.9, −2.3]	−3.6 [−9.1, +1.8]	−11.2 [−17.1, −5.2]	+25.6	61.9	64.5	61.9
Web Interaction	237	64.4	−2.9 [−6.8, +0.8]	+3.1 [−0.4, +6.5]	−28.8 [−33.8, −23.8]	+23.9	40.8	42.6	40.4
Math	120	100.0	−17.5 [−24.2, −10.8]	−26.7 [−34.2, −19.2]	−38.3 [−47.5, −30.0]	+14.4	50.8	92.5	87.5
Scientific Reasoning	150	91.3	−19.3 [−26.0, −12.7]	−21.3 [−28.7, −14.7]	−27.3 [−35.3, −19.3]	+8.3	58.7	70.7	70.7
Code Understanding	140	89.3	−16.4 [−25.0, −8.6]	−22.9 [−32.1, −12.9]	−28.6 [−37.9, −19.3]	+0.0	59.3	65.7	65.7

Table 8: **Scale and thinking gains on the medium and hard tiers and on the hard tier alone.** Gains as in Figure 6b,c (thinking: *always* minus *never*). The untiered domains are identical in both selections.

	Medium and hard tiers			Hard tier only		
Domain	Tasks	Scale	Thinking	Tasks	Scale	Thinking
Knowledge & Commons.	200	+4.4	+3.0		untiered	
Sentiment & Emotion	200	+3.4	−2.2		untiered	
Retrieval & Reranking	200	+3.9	+1.8	100	+3.4	+4.0
Intent & Routing	200	+12.0	+2.0	100	+4.9	+3.6
Safety	323	+16.2	+3.4	158	+19.8	+4.4
Task Completion Verif.	109	+17.4	+7.3	20	+3.8	−15.0
Tool & API Use	182	+25.6	+2.6	104	+26.2	+0.0
Web Interaction	237	+23.9	+1.8	100	+21.5	+2.7
Math	120	+14.4	+41.7	70	+13.6	+52.9
Scientific Reasoning	150	+8.3	+12.0	80	+11.2	+13.8
Code Understanding	140	+0.0	+6.4	70	+11.4	+1.4

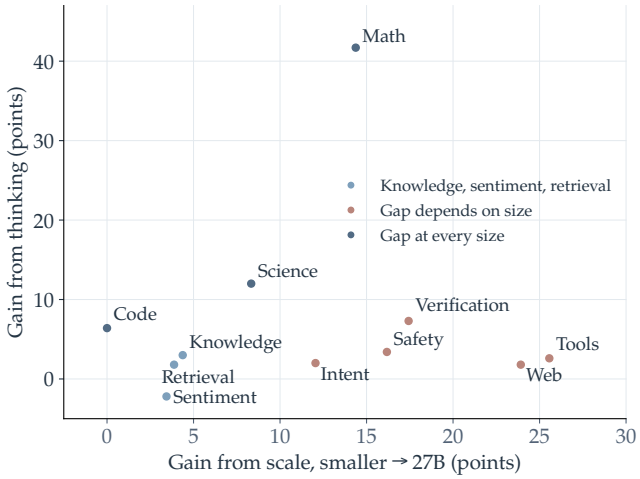


Figure 7: **Gain from scale against gain from thinking, by domain.** The bar values of Figure 6b,c, one point per domain; colors follow the three patterns of Figure 6a.